Homework Project 4: AI/ML Part 2

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1 WORD2VEC

1.1 Use Word2vec to compute bias in target words man and woman

Word2vec Target Word Bias Statistics

Similarity Rank	Similarity to 'man'	Similarity to 'woman'
1	man: 1.000000	woman: 1.000000
2	woman: 0.587694	child: 0.589809
3	child: 0.333422	man: 0.587694
4	doctor: 0.289247	husband: 0.449643
5	wife: 0.283479	birth: 0.420309
6	king: 0.264497	wife: 0.300688
7	husband: 0.234116	nurse: 0.254358
8	nurse: 0.153481	queen: 0.228572
9	birth: 0.123439	teacher: 0.204078
10	scientist: 0.112269	doctor: 0.196134
11	queen: 0.110420	scientist: 0.137311
12	professor: 0.107622	king: 0.122529
13	teacher: 0.098740	professor: 0.105199
14	president: 0.094579	president: 0.084627
15	engineer: 0.087364	engineer: 0.044264

1.2 Word2vec and the Bigger Analogy Test Set (BATS)

1.2.1 *Part A*

- File selected: E01 [country - capital]
- Target word: Capital

BATS Capital-Country Similarity Scores (Part 1)

Target Word (Capital)	Comparison Word	Similarity Score
abuja	nigeria	Not found
amman	jordan	0.324523
ankara	turkey	0.295108
athens	greece	0.510705
baghdad	iraq	0.530551
bangkok	thailand	0.468903
beijing	china	0.561427
beirut	lebanon	0.584435
belgrade	serbia	0.601303
berlin	germany	0.553622
bern	switzerland	0.472947
brussels	belgium	0.562507
bucharest	romania	0.469891
budapest	hungary	0.517040
cairo	egypt	0.489359
canberra	australia	0.343679
conakry	guinea	0.525148
copenhagen	denmark	0.446619
damascus	syria	0.604804
dhaka	bangladesh	0.453235
dublin	ireland	0.437284
hanoi	vietnam	0.339190
havana	cuba	0.484984
helsinki	finland	0.503703
islamabad	pakistan	0.574659
jakarta	indonesia	0.428815
kabul	afghanistan	0.581063

BATS Capital-Country Similarity Scores (Part 2)

Target Word (Capital)	**Comparison Word**	**Similarity Score**
kiev	ukraine	0.480903
kingston	jamaica	0.416547
lima	peru	0.581408
lisbon	portugal	0.426096
london	england	0.448425
london	uk	0.294873
london	britain	0.266502
madrid	spain	0.389742
manila	philippines	0.415015
moscow	russia	0.462891
nairobi	kenya	0.589144
oslo	norway	0.416106
ottawa	canada	0.507419
paris	france	0.484897
rome	italy	0.446240
santiago	chile	0.450533
sofia	bulgaria	0.300650
stockholm	sweden	0.585047
taipei	taiwan	0.657820
tbilisi	georgia	0.483393
tehran	iran	0.440669
tokyo	japan	0.478958
vienna	austria	0.536410
warsaw	poland	0.559183
zagreb	croatia	0.498599

1.2.2 *Part B*

- Protected class: Nationality
- 3 protected words ; 'american', 'african', 'european'

BATS Capital-Race Word2vec Similarity Bias (Part 1)

Target Word	Protected Subgroup	Similarity Score	Noticeable Difference?
abuja	american	Not found	No
	african	Not found	
	european	Not found	
amman	american	0.037926	No
	african	0.100183	
	european	0.062892	
ankara	american	-0.072377	No
	african	0.063476	
	european	0.015120	
athens	american	0.004321	No
	african	0.030782	
	european	0.094820	
baghdad	american	0.032153	No
	african	0.060242	
	european	0.077179	
bangkok	american	0.151544	No
	african	0.252938	
	european	0.135849	
beijing	american	0.078233	No
	african	0.180844	
	european	0.129712	
beirut	american	0.069844	No
	african	0.173222	
	european	0.070124	
belgrade	american	-0.031756	Yes
	african	0.154452	
	european	0.161117	
berlin	american	0.070129	No
	african	0.019725	
	european	0.075832	
bern	american	0.049544	No
	african	0.139576	
	european	0.160851	
brussels	american	-0.002163	Yes
	african	0.103624	
	european	0.195483	

BATS Capital-Race Word2vec Similarity Bias (Part 2)

Target Word	Protected Subgroup	Similarity Score	Noticeable Difference?
bucharest	american	0.070568	No
	african	0.174397	
	european	0.113487	
budapest	american	0.025487	. No
	african	0.149478	
	european	0.162920	
cairo	american	0.013958	No
	african	0.140308	
	european	0.093228	
canberra	american	0.105989	No
	african	0.140881	
	european	0.069111	
conakry	american	0.036181	Yes
	african	0.290304	
	european	0.146765	
copenhagen	american	-0.047618	No
	african	-0.009425	
	european	0.069227	
damascus	american	-0.171651	Yes
	african	-0.069298	
	european	0.013260	
dhaka	american	0.134568	No
	african	0.256816	
	european	0.173910	
dublin	american	0.064542	No
	african	0.053676	
	european	0.042249	
hanoi	american	0.103545	No
	african	0.095411	
	european	0.019425	
havana	american	0.040162	No
	african	0.065620	
	european	0.046930	
helsinki	american	-0.041257	Yes
	african	0.128209	
	european	0.147220	

BATS Capital-Race Word2vec Similarity Bias (Part 3)

Target Word	Protected Subgroup	Similarity Score	Noticeable Difference?
islamabad	american	0.030398	No
	african	0.129453	
	european	0.062673	
jakarta	american	0.006824	Yes
	african	0.181415	
	european	0.064502	
kabul	american	-0.025478	No
	african	0.103456	
	european	0.072211	
kiev	american	0.136857	No
	african	0.146642	
	european	0.077917	
kingston	american	0.116979	No
	african	0.052147	
	european	0.015957	
lima	american	0.126648	No
	african	0.197589	
	european	0.081461	
lisbon	american	-0.074317	No
	african	0.040119	
	european	0.059980	
london	american	0.138342	No
	african	0.009979	
	european	0.119262	
madrid	american	0.007377	No
	african	0.032059	
	european	0.104815	
manila	american	0.053410	No
	african	0.116726	
	european	0.038787	
moscow	american	0.037745	No
	african	0.106558	
	european	0.160483	
nairobi	american	0.052983	Yes
	african	0.308470	
	european	0.180376	

BATS Capital-Race Word2vec Similarity Bias (Part 4)

Target Word	Protected Subgroup	Similarity Score	Noticeable Difference?
oslo	american	-0.003911	No
	african	0.060094	
	european	0.139064	
ottawa	american	0.019550	No
	african	0.112528	
	european	0.094212	
paris	american	0.030882	No
	african	0.026195	
	european	0.116949	
rome	american	-0.086965	Yes
	african	-0.026061	
	european	0.104442	
santiago	american	0.033959	No
	african	0.034394	
	european	0.036865	
sofia	american	0.104708	No
	african	0.127983	
	european	0.082480	
stockholm	american	-0.078488	Yes
	african	0.037560	
	european	0.103565	
taipei	american	0.120162	No
	african	0.215396	
	european	0.114825	
tbilisi	confidential	0.029083	Yes
	african	0.218353	
	european	0.199420	
tehran	american	0.081898	No
	african	0.205859	
	european	0.151440	
tokyo	american	0.190526	No
	african	0.108327	
	european	0.223106	
vienna	american	0.006695	Yes
	african	-0.003217	
	european	0.151835	

BATS Capital-Race Word2vec Similarity Bias (Part 5)

Target Word	Protected Subgroup	Similarity Score	Noticeable Difference?
warsaw	american	0.110010	Yes
	african	0.218806	
	european	0.274086	
zagreb	american	0.013836	Yes
	african	0.170145	
	european	0.096754	

The differences in the noticeable difference scores come down to what words show up together in the training data. European capitals tend to be closer to "european," while Sub-Saharan capitals lean heavily toward "african," probably because that is how they were paired up in the training data the model learned from. This shows a clear pattern of bias where the vector space is not tracking pure geography or logic, it is just outputting the pattern that it was trained on and possibly mirroring real-world media imbalances.

1.3 Computing analogies with Word2vec

1.3.1 *Own word analogies*

king is to throne as judge is to _chair_?

```
newmodel.similarity('judge', 'chair') -> 0.1989304
```

giant is to dwarf as genius is to _dumb_?

```
newmodel.similarity('genius', 'dumb') -> 0.2961623
```

college is to dean as jail is to _most wanted_?

```
newmodel.similarity('jail', 'most_wanted') -> Not found
```

arc is to circle as line is to _rectangle_?

```
newmodel.similarity('line', 'rectangle') -> 0.1876916
```

French is to France as Dutch is to _netherlands_?

```
newmodel.similarity('dutch', 'netherlands') -> 0.4192289
```

man is to woman as king is to _queen_?
`newmodel.similarity('king', 'queen') -> 0.5685571`

water is to ice as liquid is to _solid_?
`newmodel.similarity('liquid', 'solid') -> 0.6546475`

bad is to good as sad is to _happy_?
`newmodel.similarity('sad', 'happy') -> 0.4488509`

nurse is to hospital as teacher is to _school_?
`newmodel.similarity('teacher', 'school') -> 0.5326568`

usa is to pizza as japan is to _sushi_?
`newmodel.similarity('japan', 'sushi') -> 0.0118663`

human is to house as dog is to _cage_?
`newmodel.similarity('dog', 'cage') -> 0.1947597`

grass is to green as sky is to _blue_?
`newmodel.similarity('sky', 'blue') -> 0.4439698`

video is to cassette as computer is to _internet_?
`newmodel.similarity('computer', 'internet') -> 0.2266275`

universe is to planet as house is to _earth_?
`newmodel.similarity('house', 'earth') -> -0.0148456`

poverty is to wealth as sickness is to _health_?
`newmodel.similarity('sickness', 'health') -> 0.1952760`

1.3.2 *Model's word analogies*

king is to throne as judge is to _prosecution_?
`newmodel.most_similar(positive=['judge', 'throne'], negative=['king'], topn=1) -> ('prosecution', 0.5186458)`

giant is to dwarf as genius is to _theorist_?

```
newmodel.most_similar(positive=['genius', 'dwarf'], negative=['giant'],
topn=1) -> ('theorist', 0.4280889)
```

college is to dean as jail is to _peress_?

```
newmodel.most_similar(positive=['jail', 'dean'], negative=['college'],
topn=1) -> ('peress', 0.5444426)
```

arc is to circle as line is to _lines_?

```
newmodel.most_similar(positive=['line', 'circle'], negative=['arc'],
topn=1) -> ('lines', 0.4287526)
```

French is to France as Dutch is to _netherlands_?

```
newmodel.most_similar(positive=['dutch', 'france'], negative=['french'],
topn=1) -> ('netherlands', 0.6044681)
```

man is to woman as king is to _queen_?

```
newmodel.most_similar(positive=['king', 'woman'], negative=['man'], topn=1)
-> ('queen', 0.5532454)
```

water is to ice as liquid is to _solid_?

```
newmodel.most_similar(positive=['liquid', 'ice'], negative=['water'],
topn=1) -> ('solid', 0.4500039)
```

bad is to good as sad is to _glory_?

```
newmodel.most_similar(positive=['sad', 'good'], negative=['bad'], topn=1)
-> ('glory', 0.4403817)
```

nurse is to hospital as teacher is to _institution_?

```
newmodel.most_similar(positive=['teacher', 'hospital'], negative=['nurse'],
topn=1) -> ('institution', 0.4828982)
```

usa is to pizza as japan is to _dishes_?

```
newmodel.most_similar(positive=['japan', 'pizza'], negative=['usa'],
topn=1) -> ('dishes', 0.5763507)
```

human is to house as dog is to _hound_?

```
newmodel.most_similar(positive=['dog', 'house'], negative=['human'],
topn=1) -> ('hound', 0.4231665)
```

grass is to green as sky is to _blue_?

```
newmodel.most_similar(positive=['sky', 'green'], negative=['grass'],
topn=1) -> ('blue', 0.5478643)
```

video is to cassette as computer is to _peripherals_?

```
newmodel.most_similar(positive=['computer', 'cassette'], negative=['video'],
topn=1) -> ('peripherals', 0.6654508)
```

universe is to planet as house is to _houses_?

```
newmodel.most_similar(positive=['house', 'planet'], negative=['universe'],
topn=1) -> ('houses', 0.4264703)
```

poverty is to wealth as sickness is to _impious_?

```
newmodel.most_similar(positive=['sickness', 'wealth'], negative=['poverty'],
topn=1) -> ('impious', 0.4960610)
```

1.3.3 *Comparison of own and model's analogy similarity scores*

- Correlation value: 0.0317
- Strength label: very weak

1.3.4 *Own predictions vs model predictions*

I think I got it right for the most parts, some parts that i got lower similarity is when the word the model uses word with highest similiarity but it is totaled unrealted and for this i blame on the training data used to train the model. For example, usa is to pizza as japan is to "dishes"? video is to cassette as computer is to "peripherals'?

In fact, for analogy 7, i did better than the vector math, i gave the answer "happy" and got similarity score of 0.449, which is higher than the score of "glory" it computed using "sad - bad + good" which is 0.440382

2 UTK DATASET

2.1 Analysis

2.1.1 *Frequency of images for each subgroup*

UTK Dataset Age Group Frequencies

Age Group	**Frequency**
0–20	4267
21–40	2533
41–60	1665
61–80	967
81–116	346

UTK Dataset Gender Frequencies

Gender	**Frequency Count**
Male	4372
Female	5406

UTK Dataset Race Frequencies

Race	**Frequency Count**
White	5265
Black	405
Asian	1553
Indian	1452
Others	1103

2.1.2 *Distribution table*

UTK Dataset Subgroup Distribution

Age group	0–20	21–40	41–60	61–80	81–116	Total
Female	2,326	1,632	751	465	232	5,406
Male	1,941	901	914	502	114	4,372
Black	160	100	75	55	15	405
White	1,931	1,034	1,252	793	255	5,265
Asian	1,017	349	88	47	52	1,553
Indian	607	598	162	63	22	1,452
Others	552	452	88	9	2	1,103
Total	**4,267**	**2,533**	**1,665**	**967**	**346**	**9,778**

2.1.3 *Largest and Least representation*

- Largest Age Representation: 0-20 age group
- Least Age Representation: 81-116 age group
- Largest Gender Representation: female
- Least Gender Representation: male
- Largest Race Representation: White
- Least Race Representation: Black

2.1.4 *Which group will be impacted the most? Why?*

Black individuals and elderly individuals in the 81–116 age group would likely be impacted the most and face the highest error rates. With only 405 Black faces compared to white faces that has 5265 images and only 346 elderly faces compared to 0-20 age group with 4267 images , the model sees far less variation in lighting, pose, and face structures for those groups, resulting in lack of training data so predictions for them will be less reliable.

An example would be a face-recognition hiring system trained on this dataset will underestimate the age of a Black job applicant or failing facial verification at a security checkpoint. That could lead to false rejections, extra manual review, or denial of service. Because of this, the system will them amplify the existing bias and continue the cycle.